

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths *	Prac **
5	(b)	(iii)		27.6 mm ³ g ⁻¹ minute ⁻¹ = 3 marks 27.58 = 2 marks (not to 1 dp) 110.3 = 2 marks (Using diameter instead of radius) 110.32 = 1 mark (Using diameter and not to 1dp) 137.9 = 2 marks (Using 5 minutes) 13.8 = 2 marks (Using 0.5g) 13.79 = 1marks (Using 0.5g and not 1 dp) Cross sectional area of lumen. $\pi r^2 = 1.13\text{mm}^2$ Volume used = $1.13 \times 61 = 68.93\text{mm}^3$ Divide by 5 to give per minute = 13.8mm^3 per minute. x 2 to give per g = $27.6 \text{ mm}^3 \text{ g}^{-1} \text{ minute}^{-1}$		3		3	3	
	(c)	(i)		102/145 = 0.7/ 0.70/ 0.703		1		1	1	
		(ii)		Lack of water in environment (1) Produce {a <u>lot of metabolic</u> water/ a lot of water from respiration} (1)			2	2		
		(iii)		Glycogen does not need a large oxygen supply/ ORA (1) And does not need to remove a large amount of CO ₂ /water/ ORA (1)			2	2		
				Question 5 total	0	8	10	18	4	10